

Class Feature Space Reconstruction for Automatic Modulation Open Set Recognition

Jie Chen^{ID}, Shilian Zheng^{ID}, Luxin Zhang^{ID}, Keqiang Yue^{ID}, and Zhijin Zhao^{ID}

Abstract—Automatic modulation classification (AMC) is a critical technology in the field of wireless communications. However, as the wireless communication environment becomes increasingly complex, the received signal may include unknown modulation types. Traditional closed-set recognition algorithms assume that all test samples belong to known categories. Consequently, these algorithms often misclassify unknown modulation types, failing to meet practical needs. To address this issue, in this paper, we propose a Deep Learning-Based Class Feature Space Reconstruction (CFSR) method for open-set recognition of radio signals, aiming to achieve accurate classification of known categories and effective detection of unknown categories. First, CFSR sets a specific class feature space for each known class, and the corresponding class samples are fitted using the AE manifold in the class feature space, thereby overcoming the class feature overlap problem caused by compressing the intra-class features into a limited space. Then, The class space open-set loss is introduced based on the class feature space. By treating the remaining class samples except the current class as unknown samples, unknown information is effectively introduced during the training process, thereby enhancing the open set recognition ability of the model. Finally, the experimental results on multiple public datasets show that the proposed CFSR has great open-set recognition performance and outperforms the existing open-set recognition algorithms.

Index Terms—Automatic modulation classification, open set recognition, class feature space reconstruction, deep learning.

I. INTRODUCTION

WITH the widespread application of wireless communication technology, the rapid increase in wireless devices has led to the scarcity of available radio spectrum resources [1]. Automatic Modulation Classification (AMC), as a crucial component of cognitive radio technology, can classify received signals based on their modulation types in real time without requiring prior knowledge of the transmitted signal. This capability effectively optimizes the performance

of communication systems in dynamic spectrum-sharing environments. Currently, AMC is widely applied in civilian and security fields, encompassing cognitive radio [2], spectrum sensing [3], electronic countermeasures [4], and radio reconnaissance [5]. However, the rapid development of wireless communication technology has led to an increase in the number and complexity of electromagnetic signals, posing more severe challenges for AMC.

Most AMC techniques are developed based on the closed-set assumption, where the modulation types encountered during the testing phase are known a priori from the training phase [6]. However, in real-world scenarios, there is a need to identify previously unseen modulation types [2]. For example, in wireless communication systems, devices may encounter interference from newly deployed communication standards or unexpected signals from other devices that were not known during training. Detecting and classifying these unknown signals is critical for improving system security and robustness [7]. When faced with unknown modulation types, closed-set AMC methods often misclassify them as known types, leading to erroneous predictions. Consequently, it is essential to extend AMC methodologies to accommodate the open-set assumption, known as open-set recognition (OSR), to effectively address this challenge.

OSR requires classification models to accurately identify known modulation types while effectively rejecting unknown ones [8]. Traditional deep learning networks focus on discriminative features of known classes during training, allocating each known class within the entire feature space. This approach leads to unknown classes being mapped into the same feature space, causing significant overlap between the distributions of known and unknown classes [9]. To address this issue, many previous works have proposed Auto-Encoder(AE)-based methods [10], [11], [12], [13] and prototype-based methods [9], [14], [15]. These approaches aim to learn compact representations of known classes, enabling the model better to separate the feature spaces of known and unknown classes.

A prototype refers to one or more representative samples or latent features that characterize each class. Prototype-based methods, such as Generalized Prototype Learning (GCPL) [9], Augmented Reciprocal Point Learning (ARPL) [14], and Adversarial Kinematic Prototype Framework (AKPF) [15], use prototypes to represent each known class in the feature space. These methods perform both closed-set and open-set recognition by calculating the distance between samples

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and their corresponding prototypes. A graphical description is shown in Fig. 1(a). However, using a single or a limited number of points in the prototype learning framework may not sufficiently represent the entire class. Moreover, compressing intra-class features into a confined space can result in class feature overlap and the neglect of unknown class information. These issues in class representation introduce open space risk, thereby degrading classification performance.

AE-based methods, including class-conditional autoencoder [10], conditional Gaussian distribution learning [11], and class-specific semantic reconstruction (CSSR) [13], learn latent class representations by reconstructing the original input or feature input and perform open set recognition by the difference in reconstruction errors between known and unknown classes. These methods can be viewed as learning low-dimensional manifolds to fit the distribution of known samples, as illustrated in Fig. 1(b). Compared to prototype learning, AE-based methods provide enhanced capability for feature representation. However, existing AE-based methods, including CSSR, typically use a shared latent space for all known classes, which may cause overlapping regions among classes and insufficient discrimination. Moreover, a single continuous low-dimensional manifold may contain inter-class regions, resulting in known class classification errors, and multiple class-specific low-dimensional manifolds only consider the feature representation of each known class and also ignore the unknown class information.

To address these limitations, this paper proposes a Class Feature Space Reconstruction (CFSR) method for open set recognition of radio signals. First, CFSR uses a deep neural network to extract the universal feature representation of radio signals. Then, unlike CSSR, which encodes all known classes into a shared latent space, CFSR constructs an independent class-specific feature space for each known class and uses an autoencoder (AE) manifold in each space to model the class samples, overcoming the problem of inter-class feature overlap that may be caused by compressing intra-class features into a limited space. Then, CFSR further introduces a class space open-set loss function defined in the class-specific space, where samples from other known classes are treated as pseudo-unknowns during training. This strategy effectively incorporates unknown class information into the learning process and enhances the open set recognition ability of the model. The representation of its feature space is illustrated in Fig. 1(c). Compared with the existing prototype method, the CFSR proposed in this paper solves the problem of class representation overlap by learning the class feature space. Compared with the AE-based method, the CFSR proposed in this paper regards known classes as unknown and effectively improves the rejection ability of unknown classes.

In general, the main contributions of this paper are as follows:

- We propose a novel open-set recognition method called CFSR. This method defines a feature space for each class and employs an autoencoder manifold reconstruction network to map the features of each class into its corresponding feature space. CFSR establishes clear boundaries between the distributions of different classes,

effectively addressing the issue of class representation overlap.

- We propose a class space open set loss that treats samples from known classes, excluding those from the current class, as unknown samples based on class feature space. This strategy equips the model with the ability to recognize unknown samples during the training phase, thereby effectively improving the ability to reject unknown samples.
- We evaluate the proposed method under various conditions. Experimental results demonstrate that CFSR outperforms baseline methods on multiple public datasets, especially in challenging scenarios such as low SNRs and large-scale datasets.

The rest of this paper is organized as follows. In Section I, we discuss the related work. In Section III, we introduced the preliminary knowledge of open set recognition of modulated signals. In Section IV, we explain the proposed CFSR method in detail. In Section V, we evaluate the performance of the proposed method. Finally, we conclude in Section VI.

II. RELATED WORK

A. Automatic Modulation Classification

AMC is one of the key technologies in communication systems. Most traditional modulation classification algorithms extract features based on expert knowledge and then input them into a classifier for classification. For example, Swami and Sadler [16] proposed to use fourth-order cumulants in a hierarchical manner for radio signal recognition. This method is robust to carrier phase and frequency offset and can be classified under few-shot scenarios and low signal-to-noise ratios. Park et al. [17] used the characteristics of wavelet transform features that are sensitive to modulation type but insensitive to changes in signal-to-noise ratio, and input them into a support vector machine to achieve modulation classification. Xie and Wan [18] applied compressed sensing to a modulation recognition method based on high-order cyclic statistics to achieve modulation recognition under non-uniform samples. However, the above methods rely on the effectiveness of extracting features based on expert knowledge and usually face the problem of high algorithm complexity.

With the widespread use of deep learning in various fields, modulation classification algorithms based on deep learning continue to emerge based on its ability to automatically extract features. O'Shea et al. [19] proposed a 1D-CNN network based on the VGG architecture for small radio signal classification tasks. The network directly takes IQ signals as input without any expert feature extraction or other preprocessing. O'Shea et al. [20] proposed a 2D-CNN network for a large radio signal dataset, treating each signal as a single-channel image, and proved the effectiveness of deep convolutional neural networks under low signal-to-noise ratio. Xu et al. [21] proposed a multi-channel convolutional long short-term deep neural network (MCLDNN), which uses a long short-term memory layer (LSTM) to integrate convolutional features of different dimensions and combine spatial features with temporal features. Hong et al. [22] proposed an efficient structure

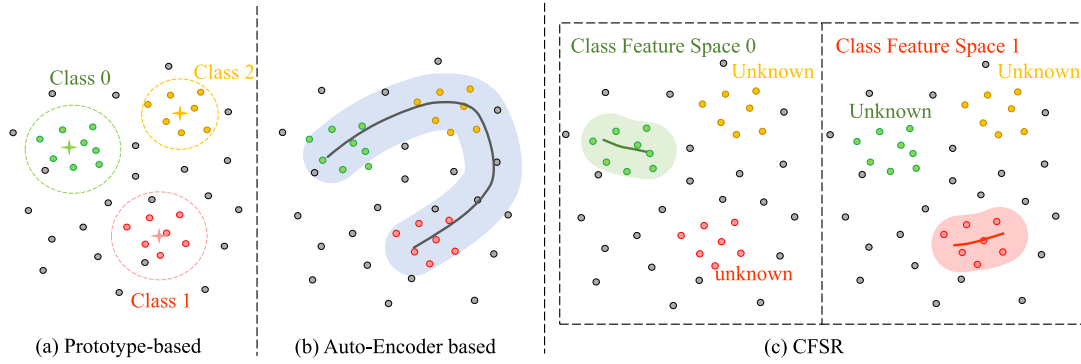


Fig. 1. Comparison of feature space representations of open set methods using prototype learning (a), Auto-Encoder (b), and the proposed method (c). Prototype learning models each class using prototype points, and AE model known samples using continuous low-dimensional manifolds. In contrast, CFSR maps each class to a distinct feature space and employs an AE to model the corresponding class within that feature space.

based on a two-layer gated recurrent unit (GRU) network, which is superior to convolutional neural networks under low signal-to-noise ratio. Chen et al. [6] converted the original IQ signal into an image through slicing and one-dimensional convolution operators as the input of the CNN network and proposed the SigNet deep learning algorithm, which has good classification performance under small sample data sets. Qiu et al. [23] integrated recurrent neural network (RNN), convolutional neural network (CNN), and graph neural network (GNN) into a single framework and proposed a hybrid heterogeneous modulation classification framework DeepSIG, which explored the modeling method for modulated signals.

The above works all focus on the classification of modulated signals under the closed set assumption, that is, the output of the neural network uses SoftMax to calculate the probability of each category, which makes it easy for the network to predict unknown samples as known. We use the automatic modulation open set recognition method to deal with this problem.

B. Open Set Recognition

The concept of OSR was first defined and a basic solution framework was proposed in [24]. Various works have developed discriminative models to tackle the OSR problem by treating unknown class detection as a binary classification issue. For example, Bendale and Boulton [25] addressed the lack of robustness in prediction probabilities of the SoftMax function by introducing the OpenMax function, which calculates Weibull distribution probabilities to explicitly obtain prediction probabilities for unknown samples. Zhou et al. [26] introduced the placeholder learning algorithm (PROSER), which predicts open-set data by learning data placeholders. This approach transforms closed-set training into open-set training, using placeholders as margins between known and unknown classes. Recently, prototype-based methods and autoencoder methods have demonstrated excellent performance in OSR tasks.

In prototype-based learning methods, Yang et al. [9] proposed the Generalized Convolutional Prototype Learning (GCPL), which uses prototypes to represent each known class and forces the features of the training set to be close to their corresponding class prototypes. GCPL increases the compactness of classes in the feature space, reducing the risk

in open space. Chen et al. [27] introduced the Reciprocal Point Learning (RPL) algorithm. Unlike prototypes, reciprocal points represent the potential representation of the out-of-class space for each category, forcing the training data to stay away from these points, thereby constraining the distribution of both known and unknown class samples in the feature space. Building on RPL, Chen et al. [14] introduced the Adversarial Reciprocal Point Learning algorithm, which uses Generative Adversarial Networks (GANs) to generate confusing samples that enhance the model's ability to distinguish between known and unknown classes. Additionally, Xia et al. [15] proposed the Kinetic Prototype Framework (KPF) based on prototype classification, introducing dynamic margin constraints to further improve the clustering compactness of known classes in the feature space. However, prototype-based methods are limited by the representational capacity of prototypes, and the unknown samples generated by GANs may differ from real unknown samples.

In AE-based methods, Oza and Patel [10] proposed the two-stage OSR model called C2AE. In this model, the encoder is first trained for closed-set recognition and then fixed, while the decoder uses Extreme Value Theory (EVT) to model reconstruction errors to determine the boundaries between known and unknown classes. Sun et al. [11] introduced Conditional Gaussian Distribution Learning based on variational autoencoders, which forces different latent features to approximate different Gaussian models. Zhang et al. [28] further extended the reconstruction capability of AE-based methods by proposing the Category Information-guided Reconstruction (CIR) framework, which integrates class-conditional vectors and mutual information constraints to improve class discrimination and enhance the model's ability to reject unknown classes. Huang et al. [13] proposed the Class-Specific Semantic Reconstruction algorithm, which replaces class prototypes with class-specific Auto-Encoders. CSSR establishes an AE manifold for each known class and uses reconstruction errors to determine class membership, effectively enhancing the representational capacity of prototypes.

The above-mentioned methods place all known class samples within the same feature space, which restricts the distribution of features for both known and unknown classes and increases the probability of feature overlap. Additionally,

they only consider known classes during training. To address these issues, we propose class feature space-specific learning.

III. PRIMILINARIES

A. Signal Model

In a wireless communication system, the goal of modulation classification is to identify the modulation type of the transmitted signal $s(n)$ based on the received signal $r(n)$. A discrete-time received signal $r(n)$ can be represented as

$$r(n) = [s(n) * h(n)]e^{j(2\pi\Delta f n + \theta_0)} + w(n), \quad (1)$$

where $h(n)$ represents the time-varying impulse response of the wireless channel, $w(n)$ denotes additive white Gaussian noise with mean of 0 and variance of σ_w^2 , Δf represents frequency offset, θ_0 represents phase offset, $*$ denotes convolution operation, and $n = 0, 1, \dots, L-1$, where L represents the length of signal.

B. Problem Definition

Modulation classification is typically defined as a multi-class problem, where the goal is to determine the modulation type C_i from the received signal $r(n)$. The modulation type index C_i belongs to $\{1, \dots, C\}$, where C denotes the number of modulation types. In close-set recognition, the modulation classes in the training and test phases are the same. In open-set recognition, the test phase includes modulation types not encountered in the training phase. Given a training dataset $D_{tr} = \{(x_1, y_1), (x_2, y_2), \dots\}$ of K known class, where $y_i \in \{1, \dots, K\}$ represents the label of sample x_i . Define the known class sample data D_k with labels $\{1, \dots, K\}$ and the unknown class sample D_u with the label $\{K+1\}$. In the test phase, the test set D_{te} , which includes both known class samples D_k and unknown class samples D_u , can be expressed as $D_{te} = D_k \cup D_u = \{x'_1, x'_2, \dots\}$ with labels $\{1, \dots, K, K+1\}$. Open-set recognition requires correctly classifying all unknown samples D_u as $K+1$ and accurately classifying the known samples D_k .

C. Prototype Learning and Auto-Encoder

In prototype learning [9], a prototype center O_k is defined for each known class k , assuming the samples of known classes are distributed around their respective prototype centers. Samples from unknown classes are expected to be far from any prototype centers. By measuring the distance between a test sample and the corresponding prototype center, we can determine whether the sample belongs to a known or unknown class. The close-set and open-set recognition can be expressed as follows:

$$p(y = k | \mathbf{x}, f_\theta, O) \propto (-\min_{O_k \in O} \|f_\theta(\mathbf{x}) - O_k\|_2^2),$$

$$p(unknown | \mathbf{x}, f_\theta, O) \propto \min_k (-\min_{O_k \in O} \|f_\theta(\mathbf{x}) - O_k\|_2^2), \quad (2)$$

where f_θ represents the feature extraction network that extracts features from the input sample $\mathbf{x}$, and O represents the set of prototype centers O_k for all known classes. However, using specific prototypes cannot fully represent the entire class.

Auto-Encoders [13] learn effective representations of samples through feature reconstruction. By using a cascaded encoder E and decoder D for each known class, input samples $\mathbf{x}$ are mapped to a low-dimensional embedded manifold F_H to fit the sample distribution. The reconstruction error $\|\mathbf{z} - D(E(\mathbf{z}))\|_2^2$ is used to measure the correlation between the latent representation $\mathbf{z}$ of input sample $\mathbf{x}$ and the embedded manifold and to determine whether the sample belongs to a known or unknown class. Close-set and open-set recognition based on AE can be expressed as:

$$p(y = k | \mathbf{z}, E, D) \propto (-\min_{\substack{E_k \in E \\ D_k \in D}} \|\mathbf{z} - D_k(E_k(\mathbf{z}))\|_2^2),$$

$$p(unknown | \mathbf{z}, E, D) \propto \min_k (-\min_{\substack{E_k \in E \\ D_k \in D}} (\|\mathbf{z} - D_k(E_k(\mathbf{z}))\|_2^2)), \quad (3)$$

where E and G represent the sets of encoders and decoders for all known classes, respectively. The autoencoder method has stronger expressive power than prototype learning by embedding the corresponding known class distribution. However, the class-specific manifold only considers the feature representation of each known class, ignoring the possible unknown class information, resulting in a decrease in open set recognition performance.

IV. METHOD

A. Overall Framework

The proposed CFSR method establishes a class feature space for each known class and employs reconstruction error as the classification criterion. Its overall framework is shown in Fig. 2, which mainly consists of three components: the backbone network B , the projection multi-layer perceptrons (MLP) $\{p_1, p_2, \dots, p_K\}$, and the class space AEs $\{AE_1, AE_2, \dots, AE_K\}$. The backbone network B is a modified one-dimensional deep residual network based on ResNet18 [29] to extract latent representations $\mathbf{z}$ from raw signals $\mathbf{x}$. The latent representations $\mathbf{z}$ extracted by the backbone network B are projected into different class feature spaces through the projection MLP p_k . In each class space AE_k , the encoder E_k and the decoder D_k are symmetric multi-layer perceptrons (MLPs) respectively. In each class feature space, the class space AE fits the corresponding known class distribution. In addition, the manifold embedding $\mathbf{z}_k^m$ generated by the class space AE is used as the input of the binary classification MLP $f_{c_k}(\cdot)$, which effectively introduces unknown information during the training process.

B. Class Feature Space Reconstruction

Prototype-based methods use one or more representative samples or latent features to represent an entire class. In contrast, AE-based methods replace prototypes with AEs, using one or more low-dimensional manifolds to fit the distribution of known class samples. Both approaches map all samples to the same feature space, partitioning the space according to the number of known classes and reserving areas for unknown class samples, as illustrated in Fig. 1(a) and 1(b). However,

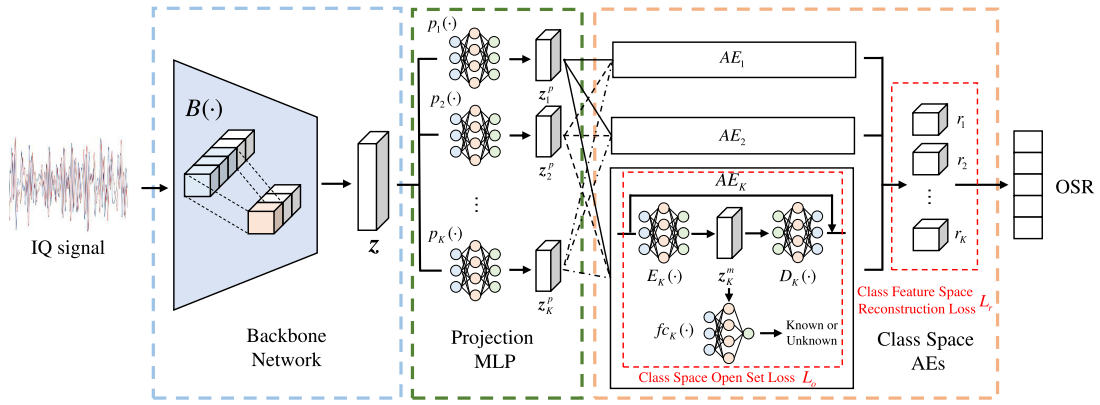


Fig. 2. Overall architecture of our proposed class feature space reconstruction method.

compressing class features into a limited space can lead to the loss of information about unknown classes. Additionally, when the number of known classes is large, the limited feature space may become overly congested, making it difficult to accurately distinguish between classes, resulting in feature overlap problems.

To address the above issues, CFSR assigns an independent feature space for each known class. First, the backbone network B extracts the latent representation z of the input x . Then, the latent representation z is mapped to the class feature space V_i through the projection MLP p_i , thereby obtaining the class space feature $z_i^p = p_i(z)$, where i represents the index of the projection MLP. Through the projection MLP, the shared features are separated into class-specific subspaces, providing cleaner and more discriminative inputs for each autoencoder to learn its own reconstruction manifold effectively. Given that AE has stronger expressive capabilities than prototypes, we use a class-space AE to fit the known classes. For the class space feature z_i^p , the reconstructed output is $\hat{z}_{i,j}^p = AE_j(z_i^p)$, where i represents the index of the known class. Then, the reconstruction error is calculated by minimizing the $L1$ norm, which can be expressed as:

$$e_{i,j} = d(z; p_i, AE_j) = \|p_i(z) - AE_j(p_i(z))\|_1. \quad (4)$$

This gives the reconstruction error of the sample x within the class feature space V_k for the class space AE_i . By considering the reconstruction errors of x in each class-space AE across different class feature spaces, a $K \times K$ error matrix Er can be formed, which can be expressed as:

$$Er = \begin{pmatrix} e_{1,1} & \cdots & e_{1,K} \\ \vdots & \ddots & \vdots \\ e_{K,1} & \cdots & e_{K,K} \end{pmatrix}. \quad (5)$$

In this error matrix Er , the i -th column represents the reconstruction error of class space AE_i across all class feature spaces, while the k -th row indicates the reconstruction error for each class space AE within the class feature space V_k . To train a robust projection MLP, the goal is to minimize the reconstruction error of input x from class k within feature space V_k across all class spaces AE . This means that for each column $Er_{:,j}$ in the error matrix, the error $Er_{k,j}$ should be minimized. The softmax function is used to compare the reconstruction

errors of the sample across different class feature spaces. For class space AE_i , the probability that sample x belongs to the class feature space V_k can be expressed as:

$$p(y = k | z, AE_i) = \frac{\exp(-d(z; p_k, AE_i))}{\sum_{m=1}^K \exp(-d(z; p_m, AE_i))}, \quad (6)$$

where K denotes the total number of known classes. Finally, by maximizing the log-likelihood to train the network, the class feature space reconstruction loss L_r can be obtained:

$$L_r = -\frac{1}{M} \sum_{j=1}^M \sum_{i=1}^K \sum_{k=1}^K \log p(y = k | B(x_j), AE_i), \quad (7)$$

where M represents the batch size and $B(\cdot)$ represents the backbone network used to extract latent representations z from the input sample x .

C. Class Space Open Set Loss

In the feature space, the prototype-based and AE-based methods reserve regions for unknown samples. The prototype learning method aims to confine known class samples within a limited space as much as possible, while the AE-based method reduces the occupation of unknown regions by learning low-dimensional manifolds with stronger feature representations. However, both methods indirectly enhance open set recognition performance by constraining known class samples, without considering unknown class samples during training. This oversight leads to a neglect of unknown class information. Some approaches attempt to generate new samples using Generative Adversarial Networks to indirectly simulate the distribution of unknown class samples [14], [30]. However, these methods infer the distribution of unknown samples based on the distribution of known class samples in the feature space, which means the generated samples may not represent true unknown samples. Additionally, generating new samples with GANs requires extra computational resources, which introduces additional computational overhead.

To address the above problems, CFSR introduces class perspective changes and regards samples of other classes except the current class as unknown samples, thereby introducing unknown samples into the feature space of each class during the training process, so that the boundary between known

Algorithm 1 CFSR Algorithm for OSR Task

Input: Training data $D_{tr} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots\}$, hyperparameter α in loss Eq. 11, learning rate lr .

Output: The optimal backbone network $B(\cdot)$, projection MLPs $\{p_i\}_{i=1}^K$ and class space AEs $\{AE_i\}_{i=1}^K$.

- 1: Initializing the parameters $\theta_B, \theta_p, \theta_{AE}$ of the backbone network $B(\cdot)$, projection MLPs $\{p_i\}_{i=1}^K$ and class space AEs $\{AE_i\}_{i=1}^K$;
- 2: **for** $epoch = 1$ to $Epoch$ **do**
- 3: **for** $batch$ in D_{tr} **do**
- 4: Calculate the class feature space reconstruction loss L_r by Eq. 7;
- 5: Calculate the class space open loss L_o by Eq. 9;
- 6: Calculate overall loss $L_{cfsr} = L_r + \alpha L_o$;
- 7: Compute the backpropagation error and update the network.
- 8: **end for**
- 9: **end for**

classes and unknown classes can be better learned. Assume we have a well-trained backbone network B , a projection MLP p_k , and a class space AE_k . For an input $\mathbf{x}$ of class label k , the class space feature $\mathbf{z}_k^p$ is obtained through the backbone network B and the projection MLP p_k . For the class space AE_k , $\mathbf{z}_k^p$ is considered a known sample, whereas for an input $\hat{\mathbf{x}}$ not belonging to class k , AE_k considers it an unknown sample. Utilizing a binary classification MLP $f_{c_k}(\cdot)$, AE_k can distinguish between known and unknown samples. The binary classification labels are:

$$\hat{y} = \begin{cases} 1, & y = k \\ 0, & y \neq k \end{cases}. \quad (8)$$

The class space feature $\mathbf{z}_k^p$ of input $\mathbf{x}$ are fed into the encoder of AE_k , resulting in the manifold embeddings $\mathbf{z}_k^m = E_k(\mathbf{z}_k^p)$. Using the manifold embedding $\mathbf{z}_k^m$ as the input to the binary classifier $f_{c_k}(\cdot)$, we obtain the probability that the sample belongs to a known class. Consequently, the class space open set loss L_o can be defined as:

$$L_o = -\frac{1}{MK} \sum_{i=1}^M \sum_{j=1}^K [\hat{y}_{ij} \log(f_{c_j}(\mathbf{z}_{ij}^m)) + (1 - \hat{y}_{ij}) \log(1 - f_{c_j}(\mathbf{z}_{ij}^m))], \quad (9)$$

where M represents the batch size, $\hat{y}_{ij}$ represents the i -th binary classification label in the batch for the class space AE_j , and $\mathbf{z}_{ij}^m$ represents the manifold embedding obtained by the input $\mathbf{x}_i$ through the class space AE_j , which can be expressed as:

$$\mathbf{z}_{ij}^m = E_j(p_j(B(\mathbf{x}_i))), \quad (10)$$

where $E_j(\cdot)$ represents the encoder of the class space AE_k and $p_j(\cdot)$ represents the projected MLP for class j .

D. Training and Inference Procedure

The training procedure of the proposed CFSR method involves optimizing a combined loss function that incorporates

both the class feature space reconstruction loss L_r and the class space open set loss L_o . The overall training loss can be expressed as:

$$L_{cfsr} = L_r + \alpha L_o, \quad (11)$$

where α is a hyperparameter, and the summary of the CFSR algorithm is shown in Algorithm 1.

During the inference phase, the proposed CFSR method determines the class of a sample by comparing the reconstruction errors from corresponding class-space AE across different class feature spaces. Specifically, for the input sample $\mathbf{x}$, the reconstruction error $\{e_{k,k}\}_{k=1}^K$ is calculated for each class projected MLP p_k and the corresponding class space AE_k . These reconstruction errors correspond to the diagonal elements of the error matrix Er as defined in Eq. 5. Then, the class of the input sample $\mathbf{x}$ is determined to be the one with the minimum reconstruction error. The probability that input $\mathbf{x}$ belongs to a known class k can be expressed as:

$$p(y = k | \mathbf{x}, AEs) = \frac{\exp(-d(p_k(B(\mathbf{x})), AE_k))}{\sum_{i=1}^K \exp(-d(p_i(B(\mathbf{x})), AE_i))}. \quad (12)$$

To achieve open set recognition, the reconstruction error is compared with the corresponding class threshold to determine if it belongs to an unknown class. For threshold selection, we set $\eta\%$ of the samples in the training set to be considered as unknown, which allows us to determine the threshold $\hat{\tau}_k$ for class k as:

$$\hat{\tau}_k = \inf \left\{ \tau : \frac{1}{|D_{tr}^k|} \sum_{\mathbf{x} \in D_{tr}^k} \mathbb{I}[d(\mathbf{x}; p_k, AE_k) < \tau] \geq \eta \right\} \quad (13)$$

where D_{tr}^k represents all samples belonging to class k in the training set, $\inf\{\cdot\}$ represents the infimum, $\mathbb{I}[\cdot]$ represents the indicator function, which equals 1 if the condition is satisfied and 0 otherwise. If the minimum reconstruction error exceeds the threshold for that class, the sample is classified as belonging to an unknown class $K + 1$. Otherwise, the sample is assigned to the class corresponding to the minimum reconstruction error. The decision rule is as follows:

$$\tilde{y} = \begin{cases} \underset{k}{\operatorname{argmax}}(p(y=k | \mathbf{x}, AEs)), & \text{if } d(\mathbf{x}; p_k, AE_k) \leq \hat{\tau}_k \\ K + 1, & \text{otherwise} \end{cases}. \quad (14)$$

V. EXPERIMENTS

A. Experiments Design

1) *Dataset*: We conducted several experiments on a public modulation recognition dataset HKDD12 to verify the performance of the proposed open-set recognition method.

The HKDD12 dataset [31] contains 12 modulation types. Each modulated signal is provided with a signal-to-noise ratio (SNR) varying from -20 dB to 20 dB, with increments of 2 dB. Each signal sample is in the form of an IQ dual-channel

TABLE I
DETAILS OF DATASET RML2016.10B AND HKDD12

Dataset	HKDD12
Number of samples	378,000
Modulations	BPSK, QPSK, 8PSK, OQPSK, 16QAM, 32QAM, 64QAM, 4PAM, 8PAM, 2FSK, 4FSK, 8FSK
SNR range	-20 ~ 20 dB, interval 2 dB
Length of signal	512
Training set: Test set	2:1

with a length of 512. For each modulation type at every SNR level, the training set includes 1000 samples, while the test set contains 500 samples. The details of the dataset HKDD12 are shown in Table. I. In the open set recognition task, 7 modulation types are treated as known classes, while the remaining 5 are regarded as unknown classes.

2) *Experimental Setup*: All simulation experiments were conducted using the same learning rate strategy and environment. During training, the AdamW optimizer is employed with a cosine annealing learning rate schedule. The initial learning rate is 0.003, and the batch size is 64, with a maximum of 50 training epochs. We implemented the algorithms using the PyTorch framework, and the hardware used includes an Intel(R) Core(TM) i9-12900HX CPU and an NVIDIA GeForce RTX 3080Ti Laptop GPU.

3) *Evaluation Metrics*: Following existing open-set recognition methods [11], [14], we use the Area Under the Receiver Operating Characteristic Curve (AUROC) and the Open-Set Classification Rate (OSCR) [32] to evaluate the performance of the proposed open-set recognition algorithm. AUROC is a threshold-independent evaluation metric that assesses a model's ability to distinguish between known and unknown classes by plotting the relationship between the true positive rate and false positive rate and calculating the area under this curve. It effectively measures the model's performance across different thresholds. OSCR, on the other hand, evaluates the model's overall capability to reject unknown samples while correctly recognizing known samples by combining the correct classification rate and the false positive rate. The OSCR value typically ranges from 0 to 1, with higher values indicating better overall performance in open-set recognition tasks. To ensure a fair and unbiased evaluation of the proposed CFSR algorithm, we performed 5-fold cross-validation for all experiments. This strategy can avoid misleading conclusions caused by the fixed split between known and unknown categories. It can also more comprehensively evaluate the generalization ability of the model in different known/unknown category configurations.

B. Ablation Study

We conducted ablation experiments on the proposed algorithm to analyze the effectiveness of each component, including the class feature space reconstruction loss L_r and the class space open set loss L_o . We evaluated the performance by removing the two loss components in the proposed

algorithm. The configuration “ L_{cfsr} without L_r ” indicates the removal of the class feature space reconstruction loss, while “ L_{cfsr} without L_o ” indicates the removal of the class space open set loss.

The open set recognition results under different signal-to-noise ratios for the various loss configurations are presented in Fig. 3. When L_{cfsr} includes both L_r and L_o , the model consistently achieves the best open set recognition performance across various SNR levels in both AUROC and OSCR metrics. Specifically, after removing the class feature space reconstruction loss L_r , the performance declines, particularly evident in the OSCR metric when $\text{SNR} \geq 2$ dB and in the AUROC metric when $\text{SNR} \geq 5$ dB. This indicates that assigning a distinct feature space for each known class effectively enhances the open set recognition performance under different SNR conditions, highlighting the effectiveness of multiple feature spaces in improving the open set recognition capability. Similarly, when the class space open set loss L_o is removed, a significant drop in both AUROC and OSCR metrics is observed, further demonstrating that L_o plays a crucial role in the ability of the model to recognize unknown classes. In summary, these findings clearly establish the effectiveness and importance of these two loss components in the CFSR method.

C. Comparison With Baseline

To validate the effectiveness of the proposed CFSR method, we compared it with three representative open-set recognition methods, including Openmax [25], CROSR [12], CGDL [11], PROSER [26], ARPL [14] and CSSR [13]. Among them, Openmax is a classical open-set recognition method that recalibrates the recognition probability of each class using extreme value theory. CROSR and CGDL are both feature reconstruction-based methods but differ in the distributions they fit, employing the Weibull distribution and conditional Gaussian distribution, respectively. PROSER is a confidence-based method that learns data placeholders to calibrate softmax confidence and predict unknown samples, effectively transforming closed-set training into open-set training. ARPL is a prototype-based learning approach that uses reciprocal points to enable neural networks to distinguish unknown samples while employing adversarial boundary constraints to reduce open space risk. CSSR is an AE-based method that establishes an autoencoding manifold for each known class and determines class membership based on the reconstruction error of the autoencoder. To ensure a fair comparison, we employed a unified network architecture for all baseline algorithms and the proposed method, with all backbone networks using a modified one-dimensional deep residual network based on ResNet18 [29]. For the reconstruction-based methods CROSR and CGDL, we retained their original network architectures due to the unique design requirements inherent to their method.

The results of different methods under various SNR conditions are shown in Fig. 4. As the figure shows, all methods exhibit performance improvements as the SNR increases. However, CFSR consistently delivers superior results across all SNR levels, significantly outperforming all baseline methods.

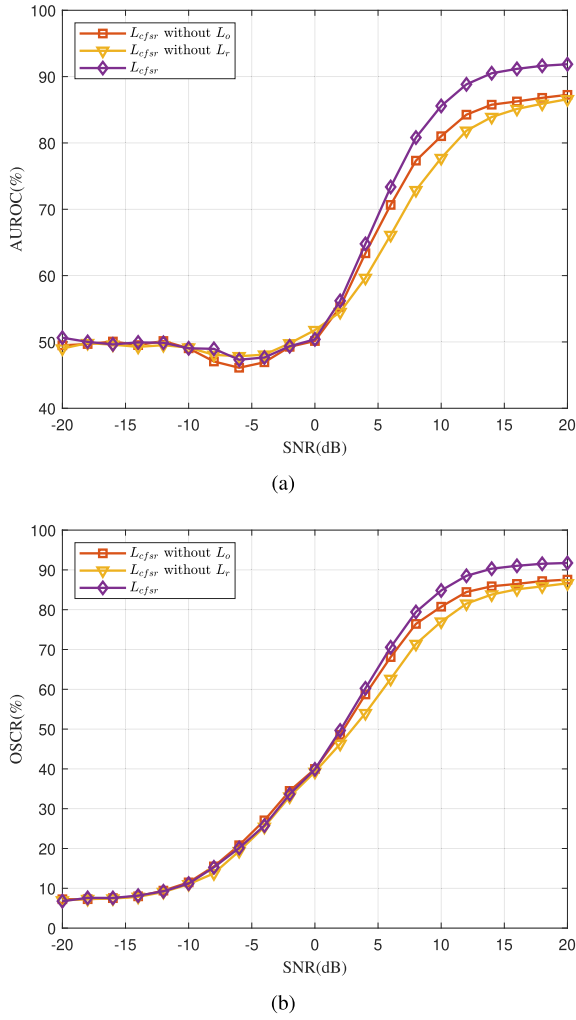


Fig. 3. Effectiveness of each loss on dataset HKDD12. (a) AUROC metrics, (b) OSCR metrics.

In particular, under high SNR conditions, CFSR achieves outstanding performance in both AUROC and OSCR metrics, reaching accuracy levels close to 90%. When the $\text{SNR} \leq 0$ dB, CFSR performs similarly to the other algorithms, reflecting the extreme challenge of open-set recognition under heavy noise conditions. But as the SNR increases, CFSR quickly surpasses the baseline methods, showing higher AUROC and OSCR scores, indicating its stronger robustness and generalization capability in open-set recognition tasks. In comparison, methods such as CSSR, CGDL, and CROSR demonstrate slightly weaker performance across the entire SNR range, while PROSER and ARPL tend to stabilize at high SNRs but still fall short of the performance achieved by CFSR. It is also noteworthy that in Fig. 4(a), many methods exhibit a noticeable performance notch at SNR around -5 dB on the AUROC metric. This phenomenon occurs because -5 dB represents a critical transition zone where the SNR reaches a threshold. The signal is no longer completely submerged in noise (at very low SNR), but has not yet reached sufficiently distinguishable (at higher SNR). This maximizes the ambiguity between signal and noise, making it particularly difficult for models to differentiate between known and unknown classes,

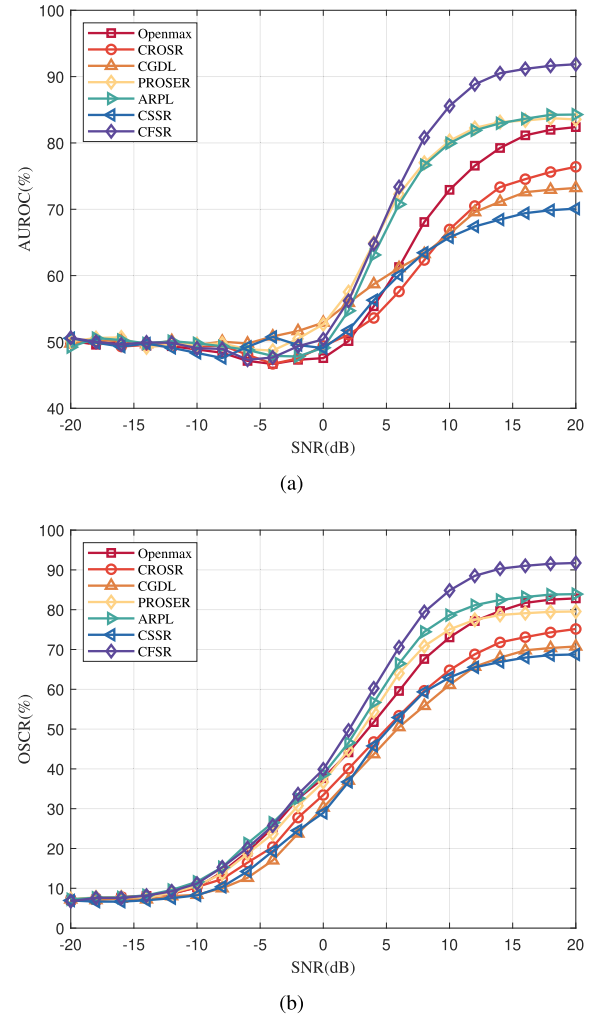


Fig. 4. The performance on dataset HKDD12 at different SNRs. (a) AUROC metrics, (b) OSCR metrics.

thus causing a temporary drop in performance. Nevertheless, CFSR shows a relatively fast recovery from this notch as the SNR increases, reflecting its stronger ability to handle noisy environments compared to the baseline methods. Overall, This result confirms the effectiveness of CFSR in handling unknown samples and validates the advantages of its proposed class feature space reconstruction loss and open-set recognition loss. The CFSR algorithm improves open-set recognition performance, especially under medium and high-SNR conditions.

D. Performance Against Openness

To validate open-set recognition performance under different levels of openness complexity, we adjust the number of known classes to modify the complexity of the open-set task. The definition of Openness [24] is as follows:

$$\text{Openness} = 1 - \sqrt{\frac{2 \times K}{2 \times K + U}}, \quad (15)$$

where K represents the number of known classes in the training set, which is the same as the number of known classes in the test set, and U represents the number of unknown classes in the test set. For the HKDD12 dataset used in this paper, the

TABLE II
AUROC AND OSCR RESULTS OF DIFFERENT METHODS AT DIFFERENT OPENNESS

Method (%)	$K = 9$ (0.074)		$K = 7$ (0.141)		$K = 5$ (0.233)		$K = 3$ (0.367)	
	AUROC	OSCR	AUROC	OSCR	AUROC	OSCR	AUROC	OSCR
Openmax [25]	71.81	71.49	79.04	79.49	76.37	72.91	69.93	69.89
CROSR [12]	64.11	60.34	72.89	71.32	51.31	48.52	53.43	53.57
CGDL [11]	66.25	64.84	70.96	67.61	66.74	51.72	60.03	54.17
PROSER [26]	78.18	77.34	82.73	78.21	78.19	71.07	72.59	72.47
ARPL [14]	75.61	75.17	82.83	82.18	78.08	74.38	73.68	73.56
CSSR [13]	69.70	66.68	68.50	66.78	74.26	69.53	70.47	70.08
CFSR (our)	89.79	89.42	89.93	89.66	80.45	76.86	74.99	74.65

TABLE III
DETAILS OF DATASET RML2016.10B AND HKDD64

Dataset	RML2016.10b	HKDD36
Number of samples	1,200,000	1,404,000
Modulations	AM-DSB, WBFM, BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, PAM4	BPSK, QPSK, 8PSK, OQPSK, 16PSK, 32PSK, 2FSK, 4FSK, 8FSK, 16QAM, 32QAM, 64QAM, 128QAM, 256QAM, 16APSK, 32APSK, 64APSK, 128APSK, 256APSK, 4PAM, 8PAM, 16PAM, MSK, GMSK, 4CPM, 8CPM, AM, FM, OOK, 4ASK, 8ASK, AM-MSK, FM-MSK, OFDM-BPSK, OFDM-QPSK, OFDM-16QAM
SNR range	-20 ~ 18 dB, interval 2 dB	-20 ~ 30 dB, interval 2 dB
Length of signal	128	1024
Training set: Test set	1:1	2:1

sum of the known classes K and unknown classes U is 12. By adjusting the value of K , the openness of the task can be modified.

As shown in Fig. 4, the OSR performance progressively improves with increasing SNR and eventually stabilizes. Therefore, to evaluate the robustness of different methods under varying openness levels, we report the average performance across the SNR range from 10 to 20 dB. This strategy can reduce the randomness caused by a specific noise level and better reflect the stability of the model under different signal conditions. The experimental results are shown in Table II. At lower openness levels ($K = 9$ and $K = 7$), all methods achieve relatively high AUROC and OSCR scores, suggesting that when the number of unknown classes is small, the models demonstrate better OSR performance. The proposed CFSR method delivers the best results at these openness levels, with AUROC and OSCR scores of 89.79% and 89.42% for $K = 9$, and 89.93% and 89.66% for $K = 7$, respectively, outperforming all other baselines. As openness increases, meaning the number of unknown classes grows, all methods show a noticeable decline in AUROC and OSCR scores, reflecting the increasing difficulty of the task. Specifically, at $K = 3$, the AUROC and OSCR of CFSR decrease to 74.99% and 74.65%, respectively, yet still outperform other baseline methods, showcasing the robustness of CFSR under higher openness conditions. In contrast, baseline methods such as PROSER and ARPL, especially CROSR and CGDL, showed more obvious performance degradation. When $K = 3$, the AUROC and OSCR of the CGDL method were only 53.43% and 53.57% respectively. Overall, CFSR has obvious

advantages at various openness levels and its performance is better than the baseline methods.

E. Performance on Large-Scale Datasets

To more comprehensively evaluate the performance of the proposed Method, we conducted experiments on the larger modulation recognition datasets including RML2016.10b [33] and HKDD36 [31], with the detail of the dataset shown in Table. III. Compared with the HKDD12 data set, the number of samples in RML2016.10b is significantly increased to verify the OSR performance of the algorithm under large-scale sample conditions. In the open set recognition task, 7 modulation types are treated as known classes, while the remaining 3 are regarded as unknown classes. In contrast, HKDD36 has significantly improved the number of modulation categories and is designed to evaluate the OSR performance in multi-category scenarios. In the open set recognition task, 26 modulation types are treated as known classes, while the remaining 10 are regarded as unknown classes. Same as the experimental setting of the HKDD12 dataset, we also performed 5-fold cross validation for all results.

The experimental results of different methods on larger-scale datasets are shown in Fig. 5 and 6. As shown in Fig.5, on the RML2016.10b dataset, all methods show improved performance with increasing SNR and tend to stabilize under high SNR conditions, but the proposed CFSR consistently outperforms classic baselines (such as Openmax and CROSR) and more advanced methods (such as CGDL, PROSER, ARPL, and CSSR), especially when $\text{SNR} \geq 10$ dB,

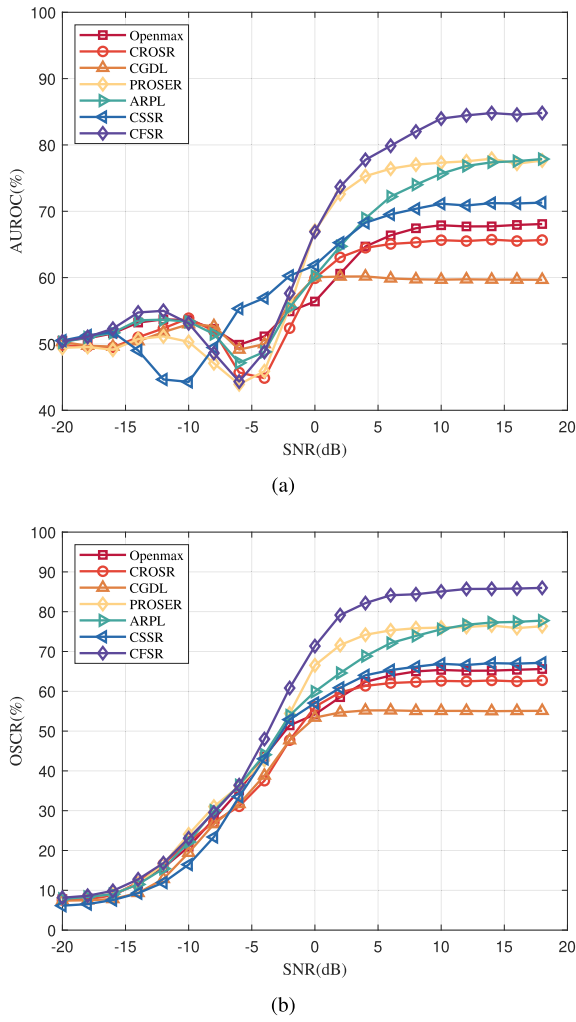


Fig. 5. The performance on dataset RML2016.10b at different SNRs. (a) AUROC metrics, (b) OSCR metrics.

with AUROC and OSCR scores of about 85% and 86%. CFSR shows a significant improvement of about 8–10% compared to PROSER and ARPL, and the improvement is more obvious compared to CSSR. It is worth noting that there is a clear performance gap between CFSR and other reconstruction-based or confidence-based methods (such as CGDL and CROSR), which have limited performance improvement under increasing SNR. This shows that CFSR are particularly effective in large-scale open set recognition tasks. In addition, in Fig. 5(a), for the AUROC metric, all methods show a significant performance drop at SNR around -5 dB, reflecting the transition region where the signal and noise ambiguity reaches its peak. However, CFSR shows a faster recovery from this gap, highlighting its stronger robustness and recovery ability. Overall, the CFSR method delivers superior performance and stronger generalization when processing large sample sets.

Fig. 6 presents the experimental results on the HKDD36 dataset. Similar to the RML2016.10b dataset, as SNR increases, the AUROC and OSCR metrics of the CFSR progressively improve and continue to outperform other methods at higher SNR levels. It is worth noting that the HKDD36

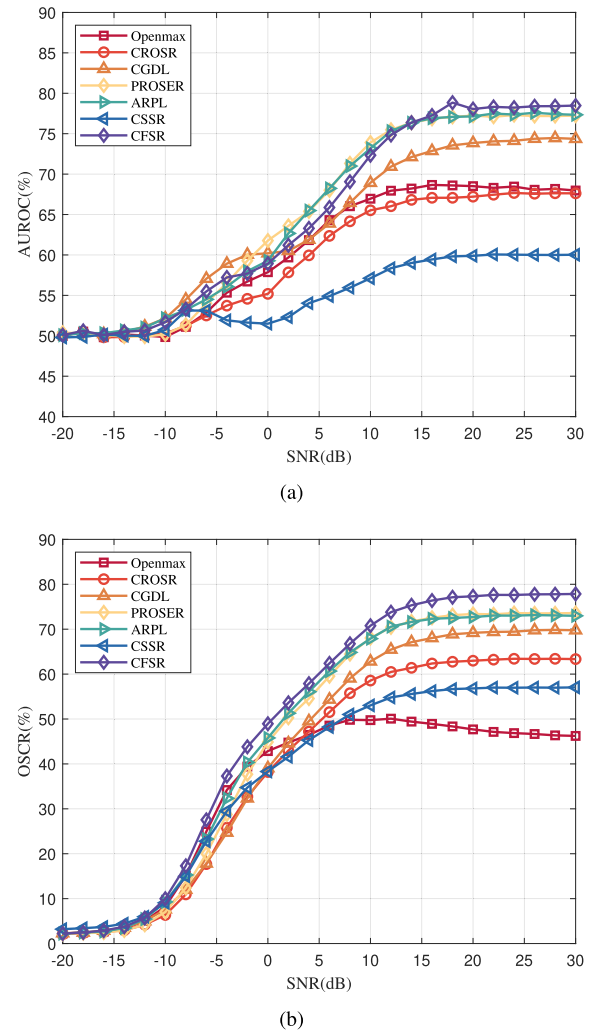


Fig. 6. The performance on dataset HKDD36 at different SNRs. (a) AUROC metrics, (b) OSCR metrics.

dataset is more complex due to the larger number of modulation categories, making the OSR task more challenging. In this case, CFSR still outperforms other methods under most SNR conditions. In particular, at higher SNR levels, CFSR achieves AUROC and OSCR scores of approximately 78% and 78%, respectively. However, we acknowledge that under certain mid-SNR conditions, the AUROC scores of CFSR do not surpass all baselines, with methods like PROSER and ARPL showing slightly better performance. This reflects the limitations of CFSR in fully separating score distributions in high-category scenarios. However, the OSCR metric, which reflects the actual performance of the decision at a fixed threshold, still shows a clear advantage in these SNR ranges, indicating that CFSR can maintain accurate classification and rejection even when the separability of the overall distribution fluctuates. Overall, while CFSR demonstrates strong performance in large-scale class scenarios, there remains room for further improvement under medium and low SNR levels. In the future, we can explore incorporating inter-class feature constraints or hybrid strategies that combine confidence-based or prototype-based mechanisms to further enhance its scalability.

VI. CONCLUSION

In this paper, we propose a Class Feature Space Reconstruction (CFSR) method for open-set recognition of modulation signals. CFSR constructs independent feature spaces for each known class, effectively resolving the issue of feature overlap between classes and improving classification accuracy. Additionally, by treating samples from all other classes as unknown during the training process, the method introduces unknown information, thereby enhancing the OSR performance. Simulation experiments show that the proposed CFSR has great open-set recognition performance and outperforms the existing open-set recognition methods.

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